BTS-III(S)-(11.20/04.21)-1685 Reg. No. | | | | | | | | |

C

B.Tech. Degree III Semester Supplementary Examination November 2020/ April 2021

CE/CS/EC/EE/IT/ME/SE- AS 15-1301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES

(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) For what value of λ, the system $x+2y=0,\ 2x+\lambda y=0$ has

(i) unique solution (ii) more than one solution.

(b) Explain eigen value and eigen vector of a matrix and state two properties of eigen values.

(c) Check whether the vectors (1,3,2), (5,-2,1), (-7,13,4) are linearly dependent or not. If dependent find the relation connecting them.

(d) Explain subspace with an example.

(e) Explain inner product space with an example.

(f) Find the Fourier series of $f(x)=x^2\ in -\pi < x < \pi$.

(g) Obtain the half range cosine series for $f(x)=x\ in\ (0,\pi)$.

(h) Find the Laplace transform of $t\,e^{-t}\cosh t$.

(i) Find the inverse Laplace transform of $\dfrac{s}{\left(s^2-a^2\right)^2}$.

(j) Find the Laplace transform of periodic function.

PART B

$(4 \times 10 = 40)$

II. (a) Explain linear system of equations and when this system has (i) unique solution (ii) more than one solution no solution.

(b) Diagonalize the matrix $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$.

OR

III. (a) Using Cayley Hamilton theorem, find the inverse of the matrix $A=\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.

(b) Find the eigen values, eigen vectors and the modal matrix of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ and hence reduce the quadratic form $x_1^2+3x_2^2+3x_3^2-2\,x_2x_3$ to a canonical form.

(P.T.O.)

IV. (a) Explain Basis and dimension of a vector space with examples. Justify your answer.

(b) Prove that intersection of any number of subspaces of a vector space V is a subspace of V.

OR

V. (a) Explain Gram Schmidt orthogonalization process.

(b) Verify whether Kernel of a linear transformation is a subspace.

VI. (a) Find the Fourier series expansion of period $2l$ for the function $f(x) = (l - x)^2$ in the range $(0,2l)$. Deduce the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

(b) Express the function $f(x) = \begin{cases} 1 & for \quad |x| \le 1 \\ 0 & for \quad |x| > 1 \end{cases}$ as a Fourier integral.

Hence evaluate $\int_0^{\infty} \frac{\sin\lambda \cos\lambda x d\lambda}{\lambda}$.

OR

VII. (a) Obtain the half range sine series of the function $f(x) = kx(x-l)$ in $0 \le x \le l$.

(b) Find the Fourier sine transform of $e^{-x}, x \ge 0$. Hence evaluate $\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx$.

VIII. (a) Using convolution theorem, find the inverse Laplace transform of $\frac{s}{(s^2+a^2)^2}$.

(b) Using Laplace transform solve $y'' - 3y' + 2y = e^{2t}$, $y(0) = -3$, $y'(0) = 5$.

OR

IX. (a) Using Laplace transform solve the integral equation $\frac{dy}{dt} + 4y + 5\int_0^t y dt = e^{-t}$ when $y(0) = 0$.

(b) Prove that $\beta(m,n) = \beta(n,m)$.

***